

Problem Set 7 (due 11:59 pm, 8 December 2025)

Remark: Students should submit executable Python programs through Canvas. Text answers in Python programs can be written in Jupyter Notebook.

1. Autocorrelation of the Absolute Return of S&P500

Download the historical data of ^GSPC from yfinance. The start date should be set to the earliest available one, which is December 30, 1927. The end date should be set to November 1, 2025 (hence the last data point will be October 31, 2025).

Suggested steps of the Python program:

- a. If the data files are exported, read the data from the data files using Pandas.
- b. The data contains non-numerical data. Clean the data using the Pandas command “dropna()”.
- c. Extract the time series of the absolute returns from the column ‘Adj Close’.
- d. Subtract the time series by its average.
- e. Plot the autocorrelation of the absolute return using the command “plt.acorr()”. The time lag should extend to around three years.
- f. For sufficiently large time lags, the autocorrelation curve becomes noisy due to fluctuations, and negative values even exist. Suggest a value of the maximum lag such that the truncated time series can be used to compute the power spectral density in Question 2.

2. Power Spectral Density of S&P500

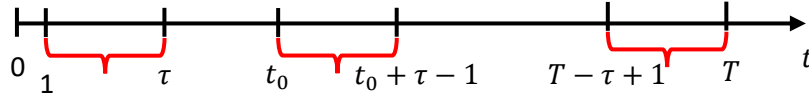
- a. Prepare the non-negative branch of the autocorrelation obtained from Question 1.
- b. Plot the power spectral density $S(f)$ of the above autocorrelation function using logarithmic scales for both axes.
- c. Suggest the crossover point between the low frequency and high frequency regimes. Obtain the values of the exponent β in the relation $S(f) \propto f^{-\beta}$ using regression (LinearRegression in scikit-learn) for both the low and high frequency regimes. However, if the high frequency data is too noisy, or the range is too short, the exponent in the high frequency regime can be omitted.

3. Detrended Fluctuation Analysis of S&P500

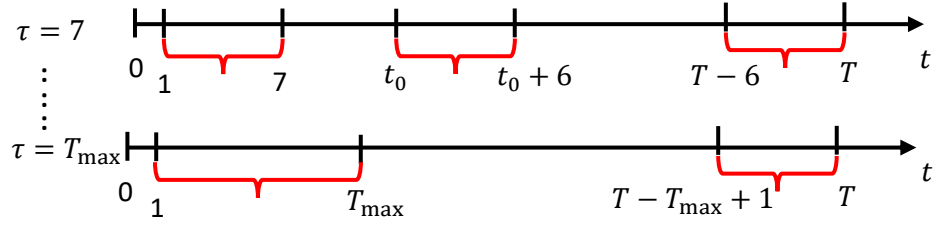
- a. Generate the series of cumulative sum.
- b. The Sliding Window Method

Next, we consider the range of time lags τ of the autocorrelation. The method of repeatedly dividing the time series into non-overlapping periods cannot generate sufficient data points for accurate determination of the exponent. So, we will use the sliding window method instead.

In the sliding window method for a time series of length T , we will compute the averaged detrended fluctuations of all data segments of duration τ within the time series. The starting instants t_0 of these segments $(t_0, t_0 + 1, \dots, t_0 + \tau - 1)$ run from 1 to $T - \tau + 1$. This provides $T - \tau + 1$ samples of data segments for analysis using the sliding window method.



For reliable analysis, let us consider data segments containing at least 7 data points. Hence, the lengths τ of the data segments run from 7 to $T_{\max}$. $T_{\max}$ is the maximum time lag suggested in Question 1. Lengths greater than $T_{\max}$ will not be considered as the autocorrelation deviates from the power law.



However, it is not necessary to increase τ linearly. As we are studying a power law, it is sufficient to increase τ exponentially. We will select the exponents x that increase from $0.1 \text{int}(10 \log_{10} 7)$ to $0.1 \text{int}(10 \log_{10} T_{\max})$ in steps of 0.1, and assign $\tau = \text{int}(10^x)$.

- c. For each segment starting at t_0 , compute the root-mean-square difference between the data points and the local trend. No fitting is required because the following result can be used directly (see Appendix):

$$\varepsilon = [\langle y^2 \rangle - \langle y \rangle^2 - m^2(\langle x^2 \rangle - \langle x \rangle^2)]^{\frac{1}{2}}, \quad \text{where} \quad m = \frac{\langle xy \rangle - \langle x \rangle \langle y \rangle}{\langle x^2 \rangle - \langle x \rangle^2}.$$

Remark: The correlation $\langle xy \rangle$ can be computed by applying `np.dot` to the arrays of x and y .

Remark: Since $\langle x \rangle$ and $\langle x^2 \rangle$ are the same for all samples, they can be computed before entering the for loop.

Remark: For $x = 0, \dots, n - 1$, $\langle x \rangle = (n - 1)/2$ and $\langle x^2 \rangle = (n - 1)(2n - 1)/6$.

- d. For each value of τ , average ε over all samples of t_0 . This yields the result $F(\tau)$. Plot $F(\tau)$ versus τ using logarithmic scales for both axes. Observe whether the power law is obeyed and whether the short time and long time behaviors are different. If they are, find the crossover time.

- e. Obtain the exponent(s) α in the relation $F(\tau) \propto \tau^\alpha$ by regression.
- f. Verify whether the relation $\beta = 2\alpha - 1$ is valid.

Appendix: The Root-Mean-Square Difference between the Data Points and the Local trend

Suppose the local trend is given by $z = mx + b$. Then the difference is given by

$$\varepsilon^2 = \langle (mx + b - y)^2 \rangle.$$

To minimize the difference, we take the derivatives with respect to m and b ,

$$\frac{\partial \varepsilon^2}{\partial m} = \langle 2x(mx + b - y) \rangle = 0.$$

$$\frac{\partial \varepsilon^2}{\partial b} = \langle 2(mx + b - y) \rangle = 0.$$

This leads to the following two equations

$$\langle x^2 \rangle m + \langle x \rangle b = \langle xy \rangle,$$

$$\langle x \rangle m + b = \langle y \rangle.$$

The solution is

$$m = \frac{\langle xy \rangle - \langle x \rangle \langle y \rangle}{\langle x^2 \rangle - \langle x \rangle^2}, \quad b = \langle y \rangle - m \langle x \rangle.$$

Substituting into the expression of ε^2 ,

$$\varepsilon^2 = m \langle x(mx + b - y) \rangle + b \langle mx + b - y \rangle - \langle y(mx + b - y) \rangle.$$

The first two terms vanish because they are expressions that have appeared in the derivatives. The last term is the remaining one. Hence,

$$\begin{aligned} \varepsilon^2 &= \langle y^2 \rangle - m \langle xy \rangle - b \langle y \rangle = \langle y^2 \rangle - \langle y \rangle^2 - \frac{(\langle xy \rangle - \langle x \rangle \langle y \rangle)^2}{\langle x^2 \rangle - \langle x \rangle^2} \\ &= \langle y^2 \rangle - \langle y \rangle^2 - m^2 (\langle x^2 \rangle - \langle x \rangle^2) \langle x^2 \rangle - \langle x \rangle^2. \end{aligned}$$